Minority Candidates and Minority Voter Turnout

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For this practical, we will analyze data from the following article:

Fraga, Bernard. (2015) "Candidates or Districts? Reevaluating the Role of Race in Voter Turnout," *American Journal of Political Science*, Vol. 60, No. 1, pp. 97–122.

Fraga assesses the theory that minority voters are more likely to vote in elections featuring co-ethnic candidates (i.e., candidates that share the same race of the minority voters).

A description of the variables is listed below:

Name	Description
`year`	Year the election was held
`state`	State in which the election was held
`district`	District in which the election was held (unique within state but not across states)
`turnout`	The proportion of the black voting-age population in a district that votes in the general election
`CVAP`	The proportion of a district's voting-age population that is black
`candidate`	Binary variable coded "1" when the election includes a black candidate; "0" when the election does not include a black candidate

Question 1

Fraga analyzes turnout data for four different racial and ethnic groups, but for this analysis we'll focus on the data for black voters. Load the `blackturnout.csv` dataset into R. Which years are included in the dataset? And how many different states are included in the dataset?

(Hints: You can use the `table()` function to get a frequency table of each year in the data; or it may be easier to use the `unique()` function just to get a list of the unique values for a given variable. Sometimes it can be difficult to count the unique values of a variable, but if you combine the `length()` function with this – `length(unique(YOUR_DATA$YOUR_VARIABLE))`, R can do it for you.)

Question 2

Next create a boxplot using the `boxplot()` function that compares turnout in elections with and without a minority (co-ethnic) candidate. The `boxplot()` function requires one argument in this format: `outcome ~ predictor`. In this case, the outcome = `turnout` and the predictor is `candidate`. Be sure to label add labels to your boxplot. Interpret what you see in the figure.

Question 3

Now let's conduct a linear regression with black turnout as your outcome variable and candidate co-ethnicity as your predictor using the `lm()` function. The `lm()` function requires two arguments in the following format: `outcome ~ predictor, data = YOUR_DATA`. For example, if my data was called `df`, the outcome (the variable for which I want to explain variation) is `dv`, and the predictor is `iv`, then you would type the following in R: `lm(dv ~ iv, data = df)`. It is generally a good idea to save your results in an object so

that you can retrieve them whenever you want. Thus, your code would be: `reg.1 <- lm(dv ~ iv, data = df)` and to see the results you would type `summary(reg.1)`.

Report the coefficient for your predictor and the intercept. Interpret the coefficients from your regression model. Based on these coefficients, what would you conclude about blacks voter turnout and co-ethnic candidates (does having a minority candidate on the ballot increase, decrease, or have no effect on turnout?)? How large or small is the coefficient. Remember that it is measured as a percentage, so .06 is a 6 percentage point increase in turnout, which could mean thousands or even millions of votes!

Question 4

Now let's say you decide to investigate the results of the previous question a bit more carefully because the elections with co-ethnic candidates may differ from the elections without co-ethnic candidates in other ways. In other words, the elections may be qualitatively different when minority candidates decide to run for office.

Create a scatter plot using the `plot()` function where the x-axis is the proportion of co-ethnic voting-age population (`CVAP`) and the y-axis is black voter turnout (`turnout`). It would be helpful to colour your points according to candidate co-ethnicity. That is, make the points for elections featuring co-ethnic candidates one color, and make the points for elections featuring non co-ethnic candidates a different color. (Hint: you can do this by using the `ifelse()` function to create a colour variable based upon the values of the `candidate` variable: `df$col <- ifelse(df$candidate == 1, "red", "blue")`, assuming that your dataframe is called `df`) Interpret the graph. Does it suggest that you may be missing an important predictor in your regression model?

Question 5

Now conduct a linear regression with black turnout as your outcome variable (`turnout`) and with candidate co-ethnicity (`candidate`) and co-ethnic voting-age population (`CVAP`) as your predictors. Recall that the general way to specify a regression in R is as follows: `reg.2 <- lm(dv ~ iv1 + iv2, data = df)`, where `dv` = your outcome, and `iv1` and `iv2` are the predictors in the `df` dataframe.

Interpret the coefficients for the two predictors, ignoring the intercept for now (you will interpret the intercept in the next question). Explain what each coefficient represents in terms of the units of the relevant variables. What does a 1-unit change in each predictor mean for your outcome (i.e., a 1-unit increase in X, is associated with a coefficient increase/decrease in Y)?

Question 6

Now interpret the intercept from the regression model with two predictors. Remember that the intercept tells you the expected value of your outcome when all predictors are set to zero (which means that `candidate = 0` *and* `CVAP = 0`). Is this intercept a substantively interesting quantity? Why or why not?